HAT-P-36b

R-1811

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_160427 · created 2026-08-03T03:30:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457505.7605 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.096 +/- 0.012
Transit depth (Rp/Rs)^2	0.91 +/- 0.24 [%]
Orbital Inclination (inc)	88.78 +/- 2.47
Airmass coefficient 1 (a1)	1.1008 +/- 0.0097
Airmass coefficient 2 (a2)	-0.0844 +/- 0.0097
Scatter in the residuals of the lightcurve fit is	0.88 %
Best Comparison Star	None
Optimal Aperture	3.69
Optimal Annulus	8.82
Transit Duration (day)	0.0935 +/- 0.0028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.096 ± 0.012 vs 0.126 (deviation 1.75σ)
Duration fit vs published	0.0935 d vs 0.0925 d (deviation 0.25σ)
Mid-time O–C	-3.4 min
Depth SNR	5.6
Residual scatter	0.88 %

Light curve

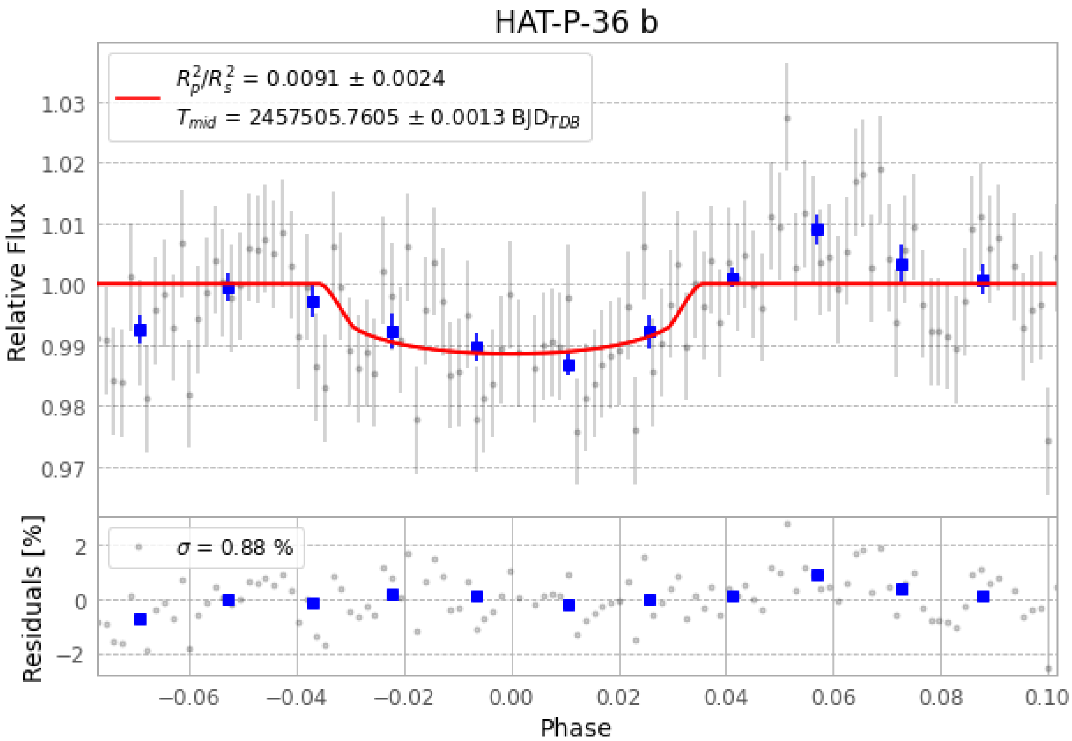

Figure 1 — FinalLightCurve_HAT-P-36 b_2016-04-26.png (SHA-256 518e12bb67a57834…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [267, 223], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 8ce54224d519d2ec…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

115 FITS frames (76 MB), `HATP-36160427034212.FITS` → `HATP-36160427092412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-160427.log (8f30500dfbaf…), FinalLightCurve_HAT-P-36 b_2016-04-26.png (518e12bb67a5…), FinalLightCurve_HAT-P-36 b_2016-04-26.csv (b7634f1f3b07…)

Compute resources

Wall time 200.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)